1. Identity

Document	STAR-PKG-STAR-GEODESY-EN · v1.5 · 2026-10-20
Package	star-geodesy
Version	0.1.1
Capability	star.geodesy.ecef
Maturity	VERIFIED
Python	⩾3.9
Source commit	94fcf7c
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_geodesy

2. Licence and obligations

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3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_geodesy-0.1.1-py3-none-any.whl
```

FILE	SHA-256
star_geodesy-0.1.1-py3-none-any.whl	09a591fb984a2bdc0589b9555c8f276acbd476924f548e56995e6ae24121b9c4
star_geodesy.py	b64fb8d6cbe9e2df7fec7f89298e3ffe4534e3ae5960dc4d050a1584739261b4
star_geodesy-0.1.1-source.zip	d68f95edd183420b3fc6f39b1ec305de6ebd64bef21956e0d5d4435e43d92619

5. What it does, and its limits

WGS-84 geodetic coordinates to and from ECEF. Standard library only.

Other datums, geoid heights, geodesic distances (see XC-003), any flight use.

- `geodetic_to_ecef(lat_deg, lon_deg, h_m)` → `(x, y, z)` in metres (WGS-84, NIMA TR8350.2 constants).
- `ecef_to_geodetic(x, y, z)` → `(lat_deg, lon_deg, h_m)`, iterative to 1e-14 rad at ANY altitude; RuntimeError if it does not converge; the forward transform of the result closes to < 1 micrometre.
- Invalid input raises `ValueError` (|lat| > 90, Earth's centre).

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_geodesy_0_1_1,
  author  = {Cavazzini, Andrea},
  title   = {{star-geodesy}: WGS-84 geodetic to/from ECEF with exact inversion at any
altitude},
```

```
  version = {0.1.1},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_geodesy/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

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Related: space activities and export control · acceptable use · evidence and limits · licence texts